Capital Bikeshare: Weather and Usage Analysis

DATA 201: Project 1

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Project Overview This analysis explores the relationship between environmental conditions and ridership patterns for the Capital Bikeshare system in Washington, D.C., using data from 2011 and 2012.

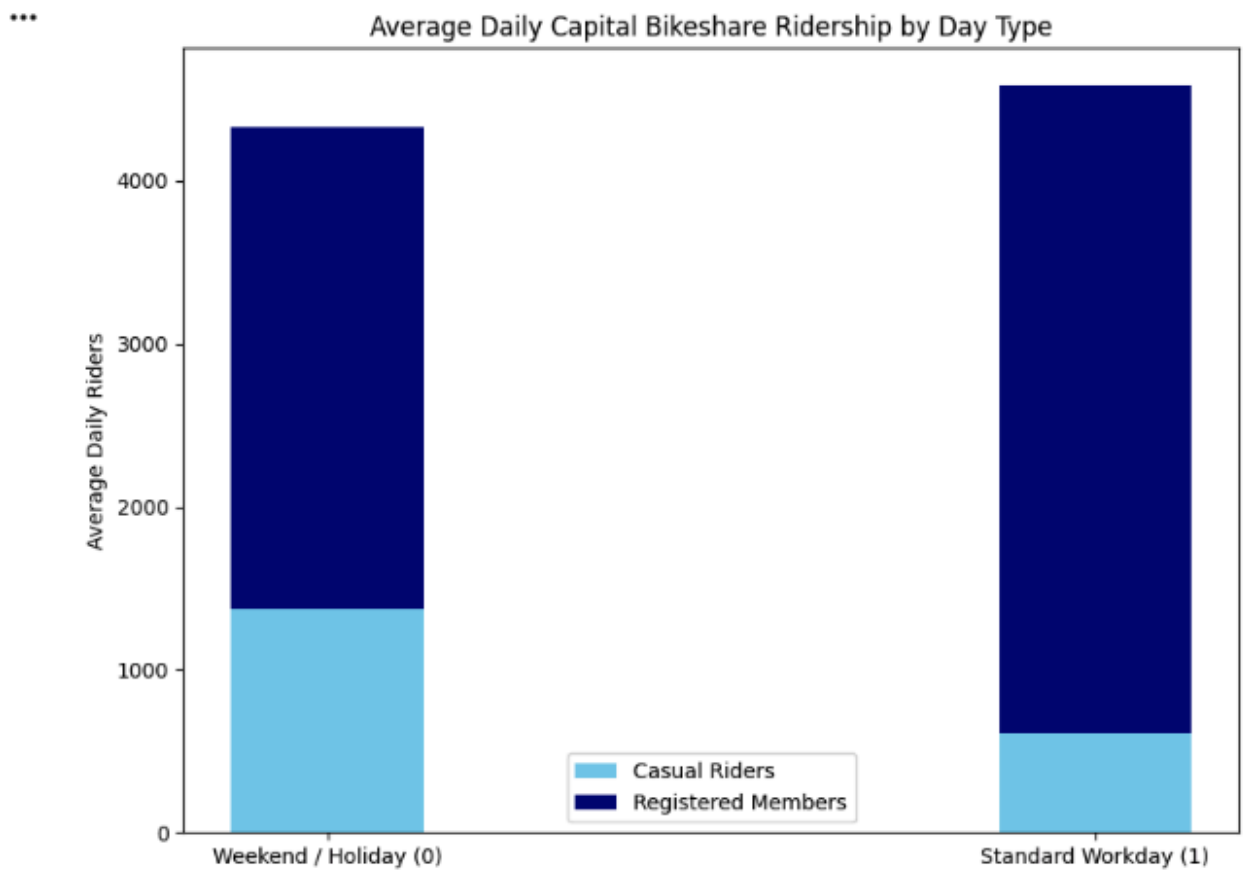
- **Data Origin:** UCI Machine Learning Repository.
- **Population:** D.C. area riders, segmented into "Casual" users (short-term passes) and "Registered" users (annual members/commuters).

- **Ethical Considerations:** The data represents successful rentals only. It does not account for "dock-out" events where a user couldn't find a bike, or the socioeconomic bias inherent in station placement which historically favors more affluent D.C. wards.

Dataset Characteristics

- **Dimensions:** 731 daily observations across 16 variables.
- **Key Variables:** `cnt` (total rentals), `casual`, `registered`, `atemp` (normalized "feels-like" temperature), and `workingday`.

User Patterns: Commuters vs. Casual Riders My analysis shows a distinct "substitution effect" based on the day of the week. On workdays, the system is dominated by registered commuters. On weekends, registered usage drops significantly, but is almost perfectly offset by an influx of casual, recreational riders, keeping the total volume (`cnt`) remarkably stable.



Temperature and Weather Deterrents

- **The Heat Threshold:** Ridership correlates positively with temperature up to a specific "comfort threshold." Once the "feels-like" temperature (atemp) reaches its D.C. summer peak, ridership plateaus and begins to decline due to extreme heat and humidity.
- **The Impact of Precipitation:** Adverse weather (Mist, Rain, or Light Snow) acts as a primary deterrent. Median ridership on clear days is significantly higher than on days with any form of precipitation, which also shows a much tighter variance in user behavior.

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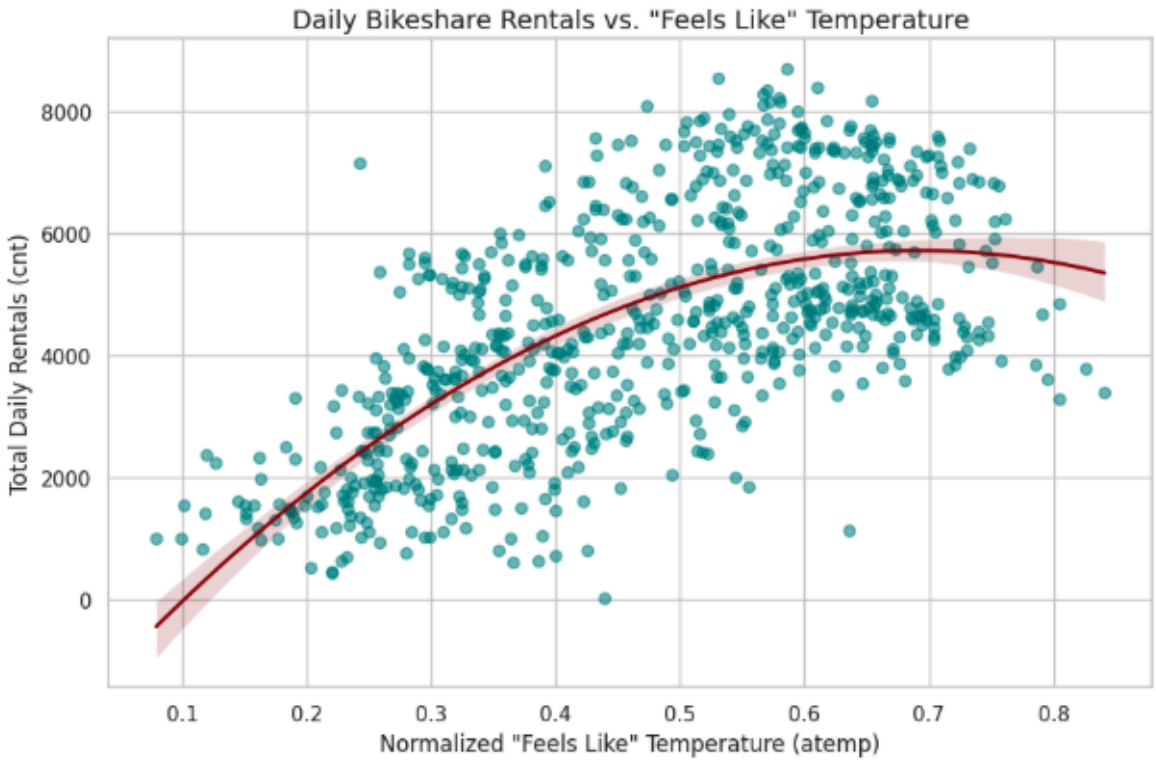

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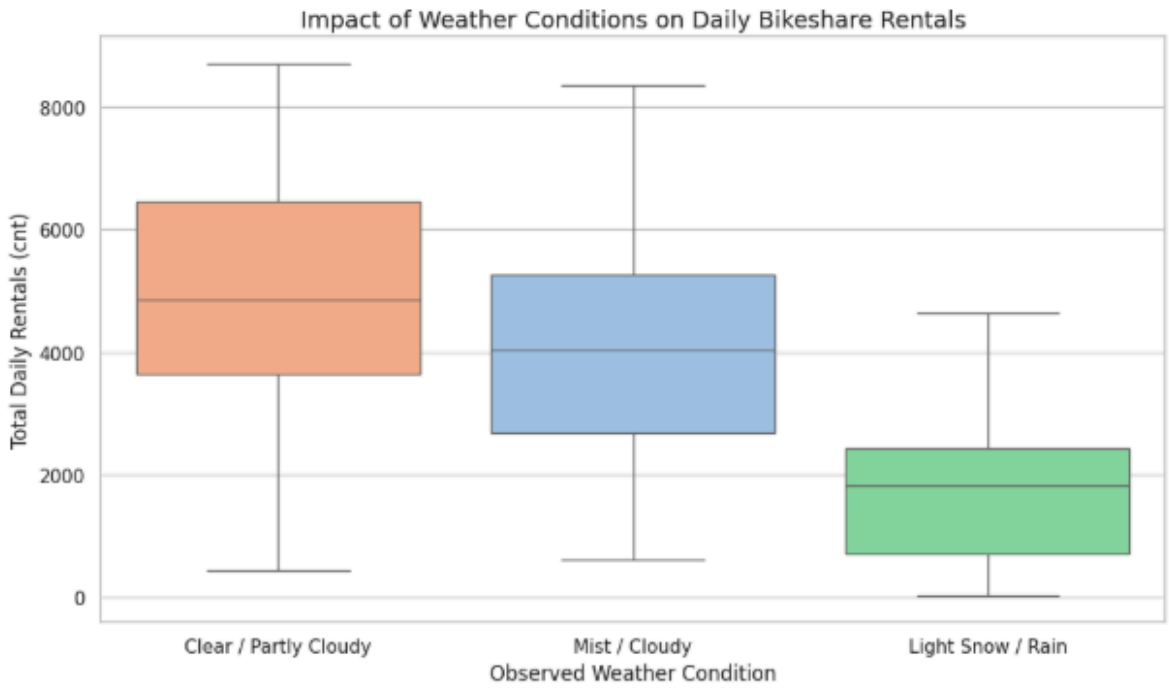
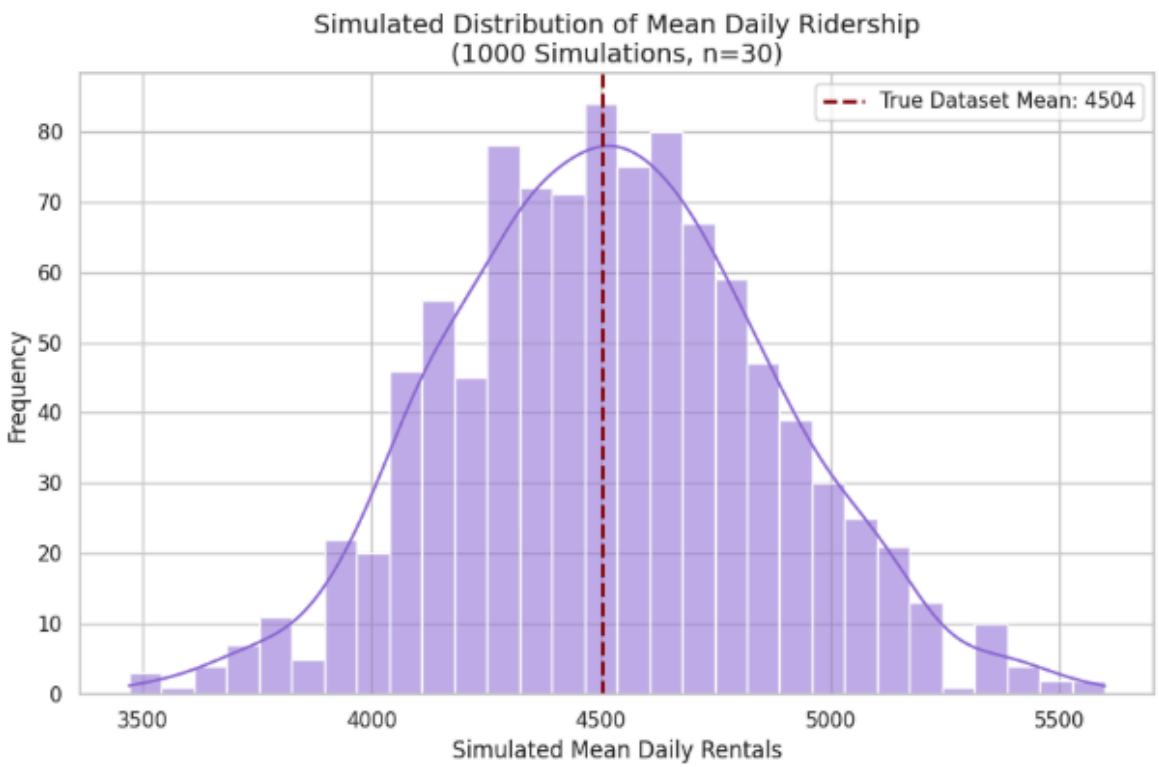


Bootstrap Simulation for Fleet Planning To account for daily volatility, I performed a 1,000-iteration bootstrap simulation of the mean daily ridership (using samples of n=30).

- **Finding:** The distribution of these sample means follows the Central Limit Theorem, clustering tightly around an average of 4,504 daily rides.
- **Logistical Application:** This provides a highly reliable baseline for long-term fleet maintenance and battery charging schedules, regardless of day-to-day weather spikes.



Final Findings

- **Volume vs. Composition:** Weather dictates *how many* people ride, while the day of the week dictates *who* is riding.
- **Practical Significance:** The `workingday` variable lacks practical significance for predicting total daily volume, as the total "fleet demand" remains consistent throughout the week.

Recommendations for Further Study Future models should utilize Multiple Linear Regression to test the interaction between humidity and temperature. Additionally, a Time-Series analysis (ARIMA) is recommended to account for the strong seasonal trends observed in the two-year period.